**Terraform Task-2 - Sudir**

**Task Description:**

**Create 2 EC2 instances on 2 different regions and install nginx using terraform script.**

**Techstacks needs to be used :**

- **AWS EC2**
- **Terraform**
- **AWS CLI**

**How do I submit my work?**

- **Push all your work files to GitHub (Code & O/P screenshot images must).**
- **Submit your URLs in the portal.**

**Terms and Conditions?**

- **You agree to not share this confidential document with anyone.**
- **You agree to open-source your code (it may even look good on your profile!). Do not mention our company's name anywhere in the code.**
- **We will never use your source code under any circumstances for any commercial purposes; this is just a basic assessment task.**

**NOTE: Any violation of Terms and conditions is strictly prohibited. You are bound to adhere to it.**

## 1. Install Terraform

On Amazon Linux:

sudo dnf  install -y terraform

```
root@ip-172-31-41-142 ~]# cat /etc/os-release
AME="Amazon Linux"
ERSION="2023"
D="amzn"
D_LIKE="fedora"
ERSION_ID="2023"
LATFORM_ID="platform:al2023"
RETTY_NAME="Amazon Linux 2023.11.20260526"
NSI_COLOR="0;33"
PE_NAME="cpe:2.3:o:amazon:amazon_linux:2023"
OME_URL="https://aws.amazon.com/linux/amazon-linux-2023/"
OCUMENTATION_URL="https://docs.aws.amazon.com/linux/"
UPPORT_URL="https://aws.amazon.com/premiumsupport/"
UG_REPORT_URL="https://github.com/amazonlinux/amazon-linux-2023"
ENDOR_NAME="AWS"
ENDOR_URL="https://aws.amazon.com/"
UPPORT_END="2029-06-30"
root@ip-172-31-41-142 ~]# sudo dnf install -y dnf-plugins-core
mazon Linux 2023 Kernel Livepatch repository
ackage dnf-plugins-core-4.3.0-13.amzn2023.0.6.noarch is already installed.
ependencies resolved.
othing to do.
omplete!
root@ip-172-31-41-142 ~]# sudo dnf config-manager --add-repo https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo
dding repo from: https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo
root@ip-172-31-41-142 ~]# sudo dnf install -y terraform
ashicorp Stable - x86_64
ependencies resolved.
================================================================================
Package                         Architecture            Version
================================================================================
nstalling:
terraform                       x86_64                  1.15.5-1
nstalling dependencies:
git                             x86_64                  2.50.1-1.amzn2023.0.1
git-core                        x86_64                  2.50.1-1.amzn2023.0.1
git-core-doc                    noarch                  2.50.1-1.amzn2023.0.1
perl-Error                      noarch                  1:0.17030-2.amzn2023.0.1
perl-File-Find                  noarch                  1.37-477.amzn2023.0.8
perl-Git                        noarch                  2.50.1-1.amzn2023.0.1
perl-TermReadKey                x86_64                  2.38-9.amzn2023.0.3
perl-lib                        x86_64                  0.65-477.amzn2023.0.8

ransaction Summary
================================================================================
nstall  9 Packages
```

terraform version

```
Installed:
  git-2.50.1-1.amzn2023.0.1.x86_64                git-core-2.50.1-1.amzn2023.0.1.x86_64
  perl-File-Find-1.37-477.amzn2023.0.8.noarch     perl-Git-2.50.1-1.amzn2023.0.1.noarch
  terraform-1.15.5-1.x86_64

Complete!
[root@ip-172-31-41-142 ~]# terraform version
Terraform v1.15.5
on linux_amd64
[root@ip-172-31-41-142 ~]# aws --version
aws-cli/2.33.15 Python/3.9.25 Linux/6.1.172-216.329.amzn2023.x86_64 source/x86_64.amzn.2023
```

---

**2. Verify AWS Access**

role is attached to the EC2:

aws sts get-caller-identity

```
[root@ip-172-31-41-142 ~]# aws sts get-caller-identity
{
    "UserId": "AROA2MRYQSGKI46762R7K:i-02466e1c625ac40cb",
    "Account": "645437362580",
    "Arn": "arn:aws:sts::645437362580:assumed-role/sudir/i-02466e1c625ac40cb"
```

---

**3. Create Project Directory**

```
[root@ip-172-31-41-142 ~]# ll
total 0
drwxr-xr-x. 3 root root 91 Jun  4 09:15 terraform-ec2
```

---

**4. Create main.tf**

```
terraform {
 required_providers {
  aws = {
   source  = "hashicorp/aws"
   version = "~> 5.0"
  }
 }
}


# Mumbai Provider
provider "aws" {
 region = "ap-south-1"
}
```

```
# Virginia Provider
provider "aws" {
 alias  = "us_east"
 region = "us-east-1"
}

# Security Group - Mumbai
resource "aws_security_group" "mumbai_sg" {
 name       = "terraform-mumbai-nginx-sg"
 description = "Allow SSH and HTTP"

 ingress {
  description = "SSH"
  from_port   = 22
  to_port     = 22
  protocol    = "tcp"
  cidr_blocks = ["0.0.0.0/0"]
 }

 ingress {
  description = "HTTP"
  from_port   = 80
  to_port     = 80
  protocol    = "tcp"
  cidr_blocks = ["0.0.0.0/0"]
 }
```

```
 egress {
  from_port   = 0
  to_port     = 0
  protocol    = "-1"
  cidr_blocks = ["0.0.0.0/0"]
 }

 tags = {
  Name = "Mumbai-Nginx-SG"
 }
}

# Security Group - Virginia
resource "aws_security_group" "virginia_sg" {
 provider    = aws.us_east
 name        = "terraform-virginia-nginx-sg"
 description = "Allow SSH and HTTP"

 ingress {
  description = "SSH"
  from_port   = 22
  to_port     = 22
  protocol    = "tcp"
  cidr_blocks = ["0.0.0.0/0"]
 }
```

```
  ingress {
    description = "HTTP"
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }

  tags = {
    Name = "Virginia-Nginx-SG"
  }
}

# Mumbai EC2
resource "aws_instance" "mumbai_ec2" {
  ami                    = "ami-0685bcc683dadb6b9"
  instance_type          = "t2.micro"
  vpc_security_group_ids = [aws_security_group.mumbai_sg.id]
```

```
 user_data = <<-EOF
           #!/bin/bash
           dnf update -y
           dnf install nginx -y
           systemctl enable nginx
           systemctl start nginx
           echo "<h1>Nginx Running - Mumbai Region</h1>" >
/usr/share/nginx/html/index.html
           EOF

 tags = {
   Name = "Terraform-Mumbai-EC2"
 }
}

# Virginia EC2
resource "aws_instance" "virginia_ec2" {
 provider            = aws.us_east
 ami                 = "ami-00e801948462f718a"
 instance_type       = "t2.micro"
 vpc_security_group_ids = [aws_security_group.virginia_sg.id]

 user_data = <<-EOF
           #!/bin/bash
           dnf update -y
```

```
    dnf install nginx -y

    systemctl enable nginx

    systemctl start nginx

    echo "<h1>Nginx Running - Virginia Region</h1>" > /usr/share/nginx/html/index.html

    EOF


  tags = {

   Name = "Terraform-Virginia-EC2"

  }

}


output "mumbai_instance_id" {

  value = aws_instance.mumbai_ec2.id

}


output "mumbai_public_ip" {

  value = aws_instance.mumbai_ec2.public_ip

}


output "virginia_instance_id" {

  value = aws_instance.virginia_ec2.id

}


output "virginia_public_ip" {

  value = aws_instance.virginia_ec2.public_ip
```

}

```
root@ip-172-31-41-142 terraform-ec2]# ll
otal 16
rw-r--r--. 1 root root  784 Jun  4 09:13 main.tf
```

---

**5. Get AMI IDs**

Mumbai:

aws ssm get-parameter \
--name /aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64 \
--region ap-south-1

**ami-0685bcc683dadb6b9**

Virginia:

aws ssm get-parameter \
--name /aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64 \
--region us-east-1

**ami-00e801948462f718a**

**Copy the AMI values into main.tf.**

---

### 6. Initialize Terraform

terraform init

```
[root@ip-172-31-41-142 terraform-ec2]# terraform init
Initializing provider plugins found in the configuration...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)

Initializing the backend...


Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

---

### 7. Validate

terraform validate

```
[root@ip-172-31-41-142 terraform-ec2]# terraform validate
Success! The configuration is valid.
```

---

### 8. Review Changes

terraform plan
also you can use terraform plan -out ec2 -> to save the plan

```
Plan: 4 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + mumbai_instance_id   = (known after apply)
  + mumbai_public_ip     = (known after apply)
  + virginia_instance_id = (known after apply)
  + virginia_public_ip   = (known after apply)

Saved the plan to: nginxtpl

To perform exactly these actions, run the following command to apply:
    terraform apply "nginxtpl"
[root@ip-172-31-41-142 terraform-ec2]#
```

---

### 9. Launch EC2 Instances

terraform apply

```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_security_group.mumbai_sg: Creating...
aws_security_group.virginia_sg: Creating...
aws_security_group.mumbai_sg: Creation complete after 2s [id=sg-010e75c1f51a73d03]
aws_instance.mumbai_ec2: Creating...
aws_security_group.virginia_sg: Creation complete after 5s [id=sg-01c95750c34062543]
aws_instance.virginia_ec2: Creating...
aws_instance.mumbai_ec2: Still creating... [00m10s elapsed]
aws_instance.mumbai_ec2: Creation complete after 12s [id=i-0b50695069f6c7536]
aws_instance.virginia_ec2: Still creating... [00m10s elapsed]
aws_instance.virginia_ec2: Still creating... [00m20s elapsed]
aws_instance.virginia_ec2: Still creating... [00m30s elapsed]
aws_instance.virginia_ec2: Creation complete after 35s [id=i-0c4d4bfd616ac8d77]

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.

Outputs:

mumbai_instance_id = "i-0b50695069f6c7536"
mumbai_public_ip = "3.7.252.218"
virginia_instance_id = "i-0c4d4bfd616ac8d77"
virginia_public_ip = "54.235.24.184"
[root@ip-172-31-41-142 terraform-ec2]#
```

---

### 10. Verify EC2 Instances

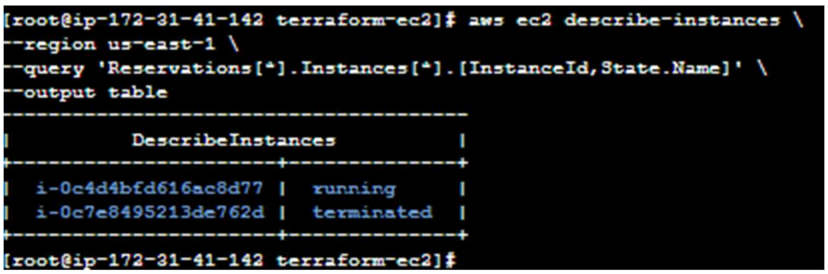
Mumbai:

aws ec2 describe-instances \
--region ap-south-1 \
--query 'Reservations[*].Instances[*].[InstanceId,State.Name]' \
--output table

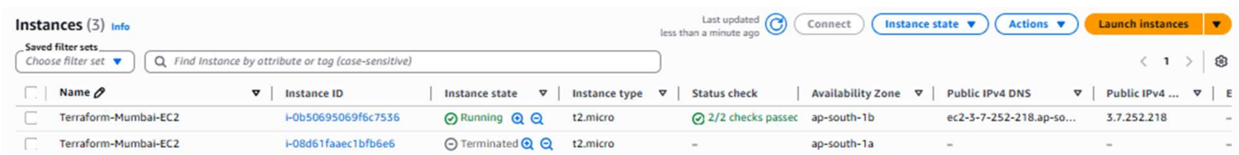
```
[root@ip-172-31-41-142 terraform-ec2]# aws ec2 describe-instances \
--region ap-south-1 \
--query 'Reservations[*].Instances[*].[InstanceId,State.Name]' \
--output table
--------------------------------------
|          DescribeInstances         |
+----------------------+-------------+
|  i-0b50695069f6c7536 |  running    |
|  i-08d61faaec1bfb6e6 |  terminated |
|  i-02466e1c625ac40cb |  running    |
+----------------------+-------------+
[root@ip-172-31-41-142 terraform-ec2]#
```

Virginia:

aws ec2 describe-instances \
--region us-east-1 \
--query 'Reservations[*].Instances[*].[InstanceId,State.Name]' \
--output table




mumbai




virginia

unable to take screenshot guvi user doesn't access

**Verify nginx on both region server**

**Mumbai**

**Virginia**


**11. Destroy**

terraform destroy

```
Changes to Outputs:
  - mumbai_instance_id   = "i-0b50695069f6c7536" -> null
  - mumbai_public_ip     = "3.7.252.218" -> null
  - virginia_instance_id = "i-0c4d4bfd616ac8d77" -> null
  - virginia_public_ip   = "54.235.24.184" -> null

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

  Enter a value: yes

aws_instance.mumbai_ec2: Destroying... [id=i-0b50695069f6c7536]
aws_instance.virginia_ec2: Destroying... [id=i-0c4d4bfd616ac8d77]
aws_instance.mumbai_ec2: Still destroying... [id=i-0b50695069f6c7536, 00m10s elapsed]
aws_instance.virginia_ec2: Still destroying... [id=i-0c4d4bfd616ac8d77, 00m10s elapsed]
aws_instance.mumbai_ec2: Destruction complete after 20s
aws_security_group.mumbai_sg: Destroying... [id=sg-010e75c1f51a73d03]
aws_security_group.mumbai_sg: Destruction complete after 0s
aws_instance.virginia_ec2: Still destroying... [id=i-0c4d4bfd616ac8d77, 00m20s elapsed]
aws_instance.virginia_ec2: Still destroying... [id=i-0c4d4bfd616ac8d77, 00m30s elapsed]
aws_instance.virginia_ec2: Destruction complete after 32s
aws_security_group.virginia_sg: Destroying... [id=sg-01c95750c34062543]
aws_security_group.virginia_sg: Destruction complete after 1s

Destroy complete! Resources: 4 destroyed.
[root@ip-172-31-41-142 terraform-ec2]#
```